- Provide the option to add a new contact if lookup is unsuccessful;
- Sync with the server when a new contact is added (or an existing one modified).

What we really need is a data structure that is specialized in these kinds of operations, we need it to support fast insertion, and at the same time to provide a way to look up an entry by value.

To be clear, when we use a plain array we don't have an efficient array's method that tells us what's the index of an element `x`, neither an efficient (as in *sublinear*) method telling us if an element is in the array or not. The only way we have to tell if an element is in the array is going through all the array's elements – although in a sorted array we could use binary search to speed up the search.

For example, we could store the strings `["the", "lazy", "fox"]` in an array, and to search for "lazy", we would skim through the whole array, element by element.

An associative array, instead, by definition has a native method that efficiently accesses the entries stored with a lookup by value. Usually this structure allows storing (key, value) pairs, for instance, we would have a list like `<("the", article), ("lazy", adjective), ("fox", noun)>`. We could search for `"lazy"` and the associative array would return `adjective`.

Another difference with regular arrays would be that the order of insertion in an associative array doesn't matter;